

CS & IT ENGINEERING

Algorithms

Greedy Method



Lecture No.- 03

By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Fractional Knapsack

Imp Terminologies

Topics to be Covered



Topic

Topic

Topic

JSD

OMP



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe

10-6 PM $12^{\text{th}}/13^{\text{th}} \rightarrow$ 8:30 - 10:30 AM

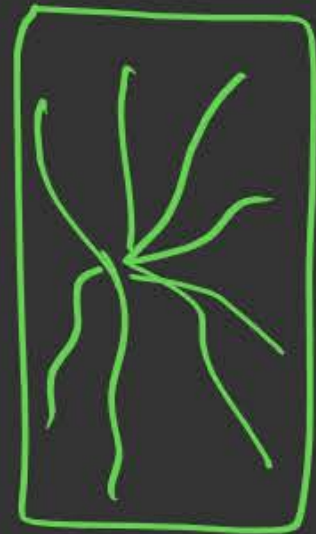
$$\frac{29}{4} = 6$$

4 AM $\rightarrow$ 6 months

4 - 8 AM

PGB/month

No Social media → no whatsapp
No movies/series/calls



3 hrs

6 months

↳ 10-12/day

6 AM

→ cf

→ 1 hr sleep

→ Dinner

11:30 PM / 12 AM

8 AM - 3 PM

"Do Your Best,
Forget the Rest"

"Quit Complaining"



Topic : Greedy Algorithm



Problem 2:

Job Sequencing with Deadline (JSD):

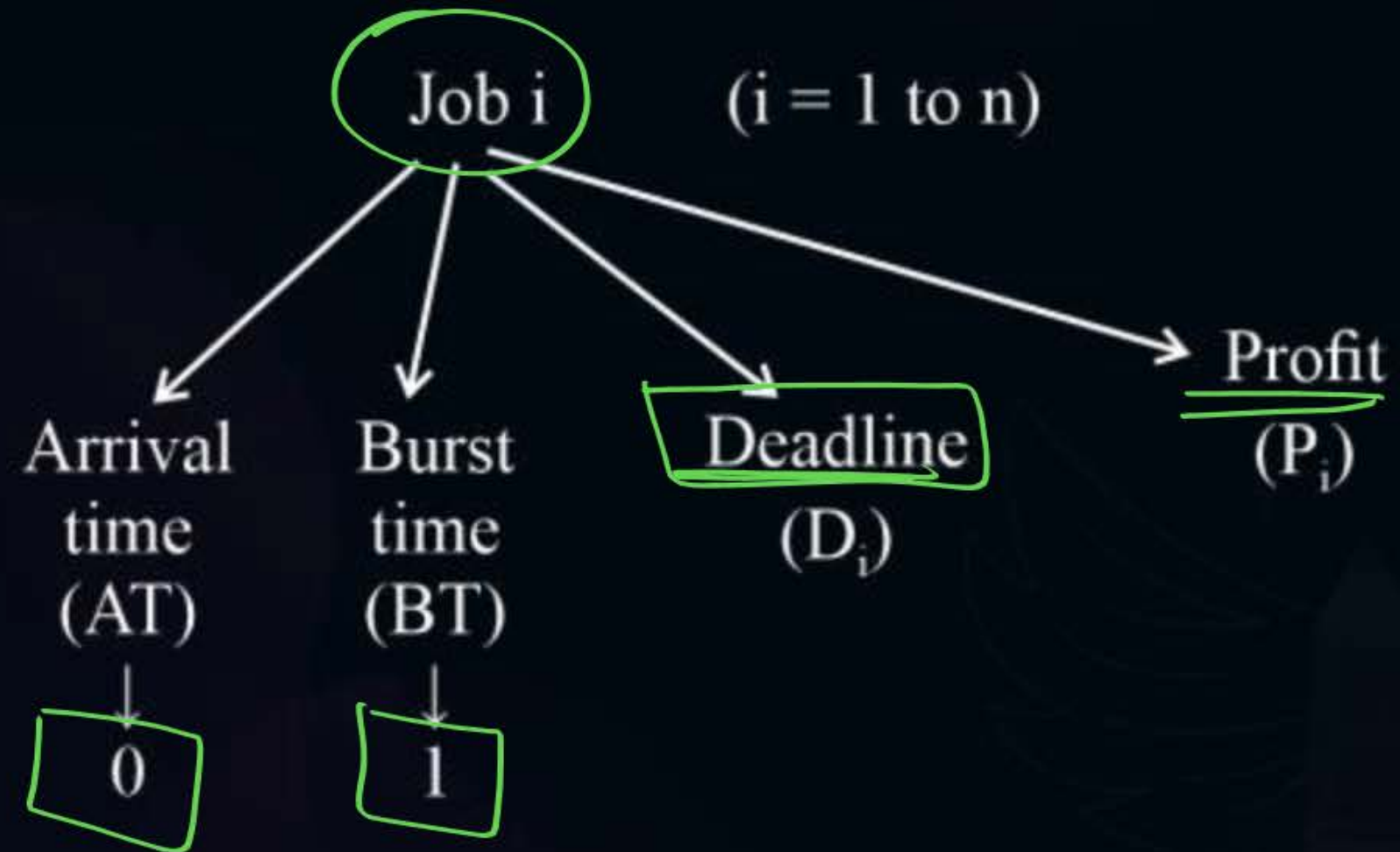
- Given a single CPU, Non-Preemptive Scheduling.
- Given n-jobs (Tasks/Processes)



Topic : Greedy Algorithm



Description:



BT=1



Topic : Greedy Algorithm



Problem Statement:

- Select a subset of 'n' given jobs, such that all the jobs in the subset are completed within their corresponding deadlines and generate the maximum profit.

Note: Only when the job gets completed with the given deadline, then only you get the associated profit (else not)

$$\text{Set} = \underline{|a| = n}$$

$$\# \text{Subsets} = \underline{\underline{2^n}}$$



Topic : Greedy Algorithm

Example:

$n = 4, J = \phi, |J| = 0$

	Profit	Deadline
J1	200	2
J2	30	1
J3	50	2
J4	80	1

$|J| = 1$

J_1

Ans: 200

$|J| = 2$

$J_4 \quad J_1 = 280$

$|J| = 3$

$J_4 \quad J_3 \quad X$



Topic : Greedy Algorithm



- This approach is Brute Force (Enumeration)

1 Jobs $\rightarrow$ Size of solution space $\rightarrow O(2^n)$

Hence, not a good idea to solve like this.

- We need a Greedy Algorithm based approach.



Topic : Greedy of Algorithms

➤ Job sequencing with deadline (JSD):

1. Enumeration logic:

$$\rightarrow O(n^2)$$

Hence,

Need greedy based Algo to reduce down the TC.



Topic : Greedy Algorithms

Eg.1. $n = 4$

Enumeration $N \rightarrow O(2^n)$	Jobs	(Di) Deadline	(Pi) Profit	
	J1	2	200	(1)
	J2	1	30	(4)
	J3	2	50	(3)
	J4	1	80	(2)

Expo

↓
Max(D) = 2

Max profit = 280



Topic : Greedy Algorithms

Greedy based optimal solution for previous Ques.

Logic:-

1. Max (Deadline) → Queue size of

J4	J1
----	----

1

x

2

x

J4	J1
----	----

1

x

2

x

2. Sort jobs on basis of decreasing profit

J1 → 2 → 200 (✓)

J4 → 1 → 80 (✓)

J3 → 2 → 50 (✗)

J2 → 1 → 30 (✗)

Max profit greedy appr.

$$= 200 + 80 + 0 + 0$$

$$= 280$$



Topic : Greedy Algorithms

#Q.

75%

28
+ 36
24
30

118

n=6	(Di) Deadline	(Pi) Profit
J1	2	28
J2	3	36
J3	4	24
J4	2	20
J5	4	16
J6	3	30

.....(3)

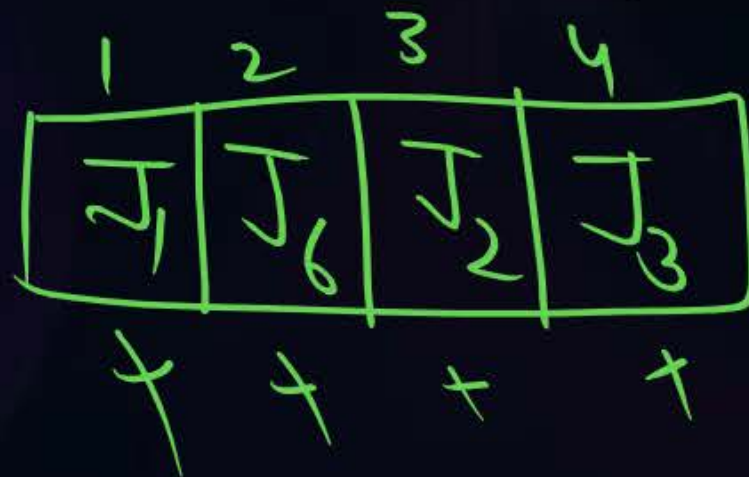
.....(1)

.....(4)

.....(5)

.....(6)

.....(2)



Max(D) = 4



Topic : Greedy Algorithms

#Q.

71%

29%

n=6	^{Profit} (Di) Deadline	^{Deadline} (Pi) Profit	
J1	56	2 ✓	(2)
J2	24	1 ✗	(5)
J3	10	5	(6)
J4	36	1 ✗	(4)
J5	60	2 ✓	(1)
J6	40	3 ✓	(3)

1	2	3	4	5
J ₁	J ₅	J ₆		J ₃

Max(D) = 5



Topic : Greedy Algorithms

#Q.

Deadline

Profit

J1

4

140

.....(2) (✓)

J2

4

170

.....(1) (✓)

J3

4

24

.....(6) (✗)

J4

4

36

.....(5) (✗)

J5

3

100

.....(4) (✓)

J6

4

120

.....(3) (✓)

J7

3

20

.....(7) (✗)

90%



90%

140
170
100
120

530 ✓

1	2	3	4
J5	J6	J1	J2

Max(D) = 4

Max profit = ?



Topic : Greedy Algorithms

1. Algorithm Greedy Job(d, J, n)
2. // J is a set of jobs that can be completed by their deadlines.
3. {
4. $J := \{1\};$
5. For $i := 2$ to n do
6. {
7. If (all jobs in $J \cup \{i\}$ can be completed
8. By their deadlines) then $J := J \cup \{i\}$
9. }
10. }



Topic : Greedy Algorithms

#Q. We are given 9 tasks T1, T2...., T9. The execution of each task requires one unit of time. We can execute one task at a time. Each task T_i has a profit P_i and a deadline d_i , profit p_i is earned if the task is completed before the end of the Deadline.

Task	T1	T2	T3	T4	T5	T6	T7	T8	T9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

1	2	3	4	5	6	7
T ₂	T ₄	T ₉	T ₅	T ₃	T ₁	T ₈



Topic : Greedy Algorithms

#Q. Are all tasks completed in the schedule that gives maximum profit?

82%

A All tasks are completed

B T1 and T3 are left out

C T1 and T6 are left out

D T4 and T6 are left out



Topic : Greedy Algorithms

#Q. What is the maximum profit earned?

93.3%

A 147

B 165

C 167

D 175

OMP



Topic : Greedy Algorithms

P3: Optimal merge patterns:- (Application of merging algo.)

Given 'n' files

It is ~~need~~^{read} to merge them using 2-way merging, such that the total number of record ^{men} moments are minimum.

Default:- [Merging 2-files at a time]



Topic : Greedy Algorithms

Concept:-

Given 2 sorted files 'A' & 'B' having 'n' and 'm' records respectively. To merge them into a single sorted file, it requires $(n + m)$ record movements.



Topic : Greedy Algorithms

Eg.2.

$n = 3$

$m = 4$

A: [2, 3, 5]

B: [1, 4, 8, 10]

[1, 2, 3, 4, 5, 8, 10,]

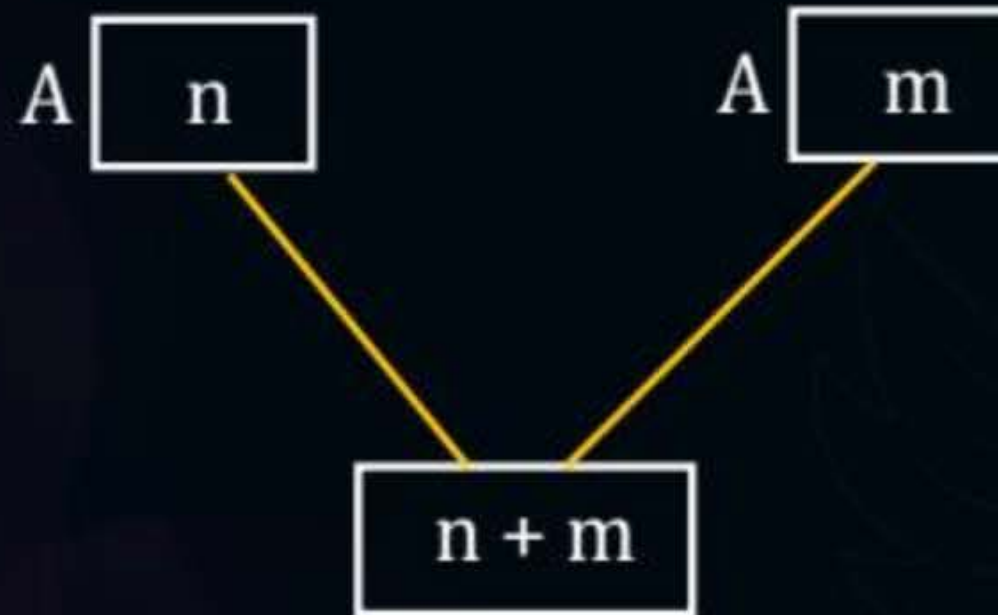
$$\begin{aligned}\text{Record movement} &= n + m \\ &= 3 + 4 \\ &= 7\end{aligned}$$



Topic : Greedy Algorithms

2-way merging Algorithm (merge sort)

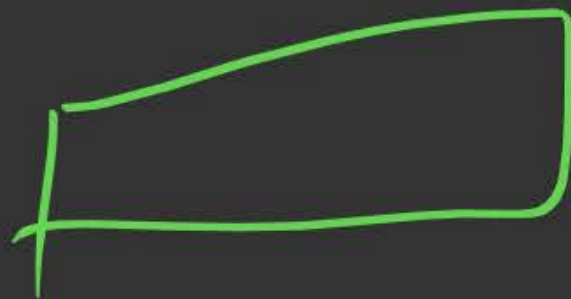
Number of elements comparisons:



Best case = $\min(n, m)$

Worst case = $(m + n - 1)$ element comparisons.

$$\begin{aligned} A &= 10 \\ B &= 15 \\ C &= 5 \end{aligned}$$



$$10 + 15 + 5$$

ABC



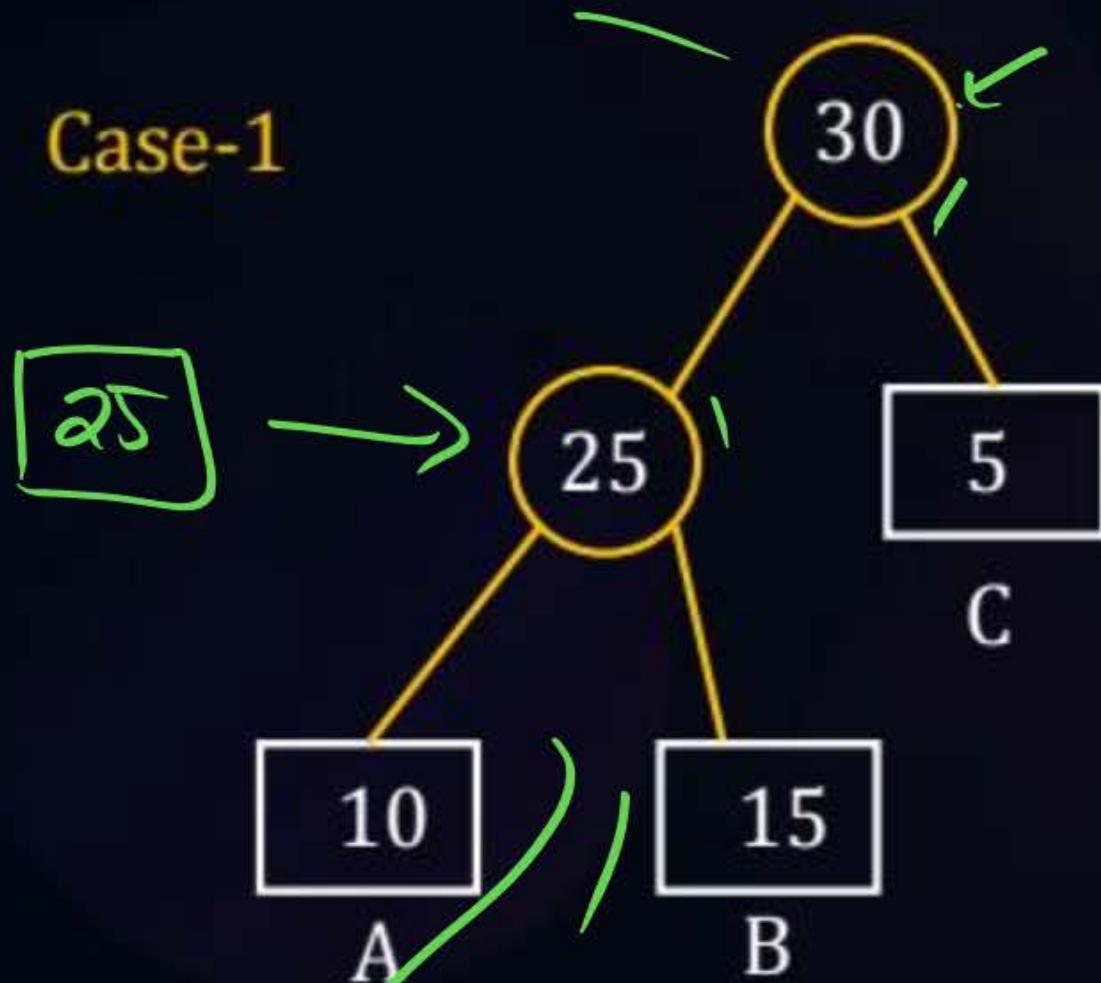
Topic : Greedy Algorithms

Eg.3. A: 10 → “number of records in files A”

B: 15 → “number of records in files B”

C: 5 → “number of records in files C”

Case-1



Total no. of records movement = $25 + 30$
= 55

Order = ABC

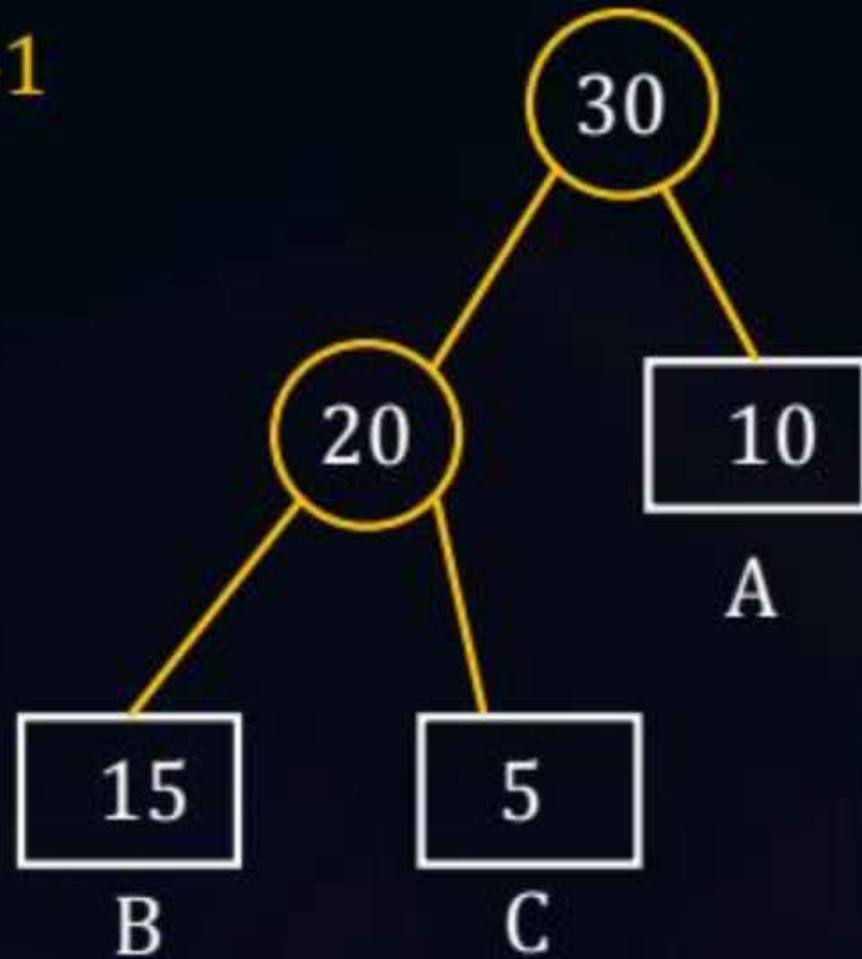
BCA



Topic : Greedy Algorithms

2-Way binary merge tree

Case-1

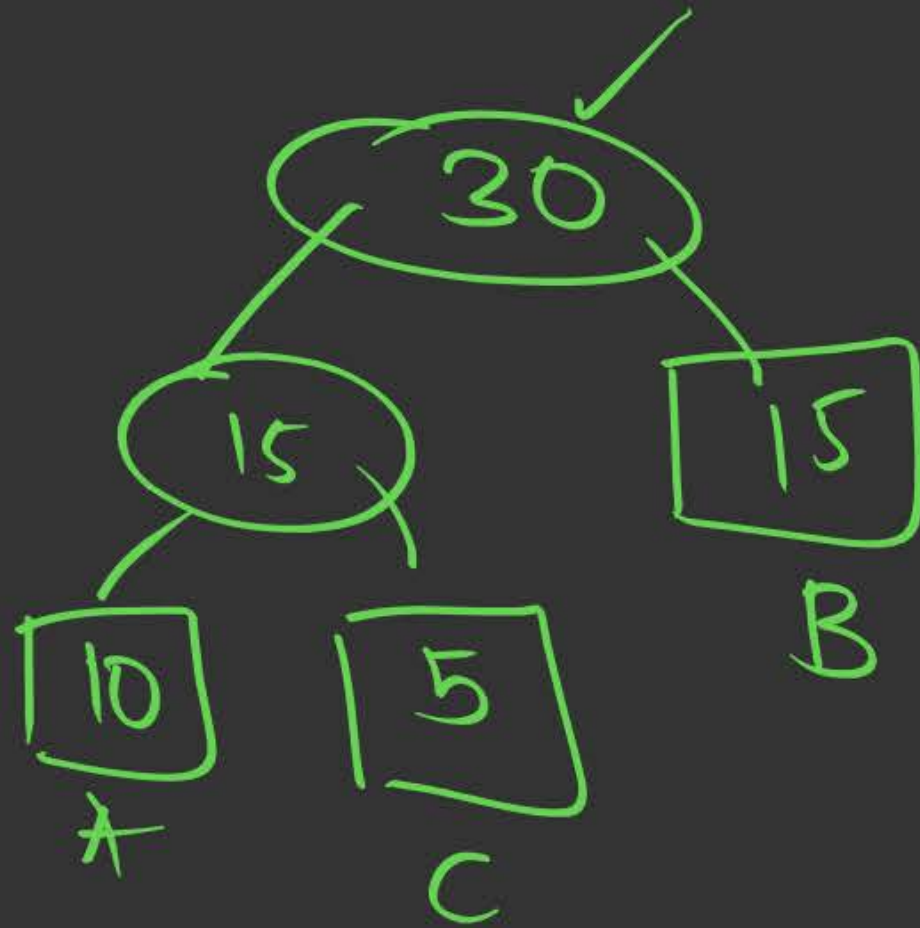


Total no. of records movement = $20 + 30$
= 50

Order = BCA

ACB

ACB



$$\text{Total RM} = 15 + 30$$

$$= \underline{\underline{45}}$$



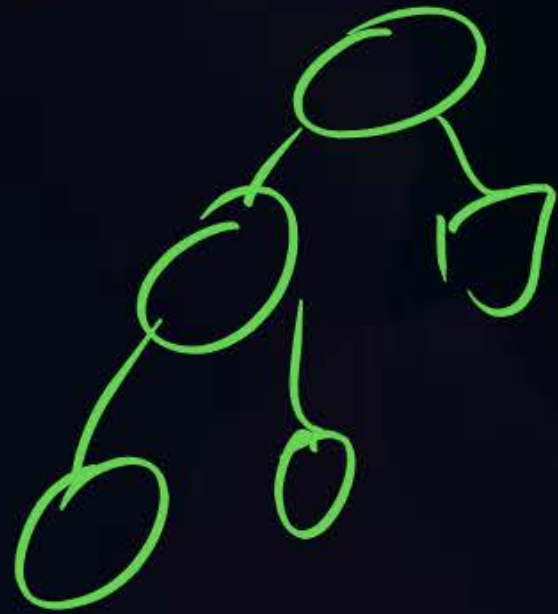
Topic : Greedy Algorithms

Observation.

Imp

? impacts

1. Ordering matters and ~~inputs~~ the total number of records movements.
2. For 'n' files, how many times 2-way merging operation is required? $\Rightarrow \underline{\underline{(n-1)}}$
3. Objective:- Minimize the total number of records movements.





Topic : Greedy Algorithms

Observation.

4. Size of solution space = ?
n! arrangement

All n! Solution are feasible.

ABC

BCA

ACB

BAC

CAB

CBA



2 mins Summary



Topic

Topic

Topic

Topic

Topic

JSD
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THANK - YOU